Md. Taifur Hossain.

CS UNDERGRADUATE · 3D VISION & ML SYSTEMS

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Final-year Computer Science undergraduate focused on memory-efficient 3D Gaussian Splatting. Published six projects on GitHub (four as tagged releases) and opened two pull requests to gsplat.

EDUCATION

International University of Business Agriculture & Technology (IUBAT) — Dhaka, BD
B.Sc. in Computer Science & Engineering (BCSE) · CGPA 3.3 / 4.00 — Expected 2027

PUBLICATIONS

Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

Designed a training-free post-processing pipeline (adaptive pruning → outlier cleaning → spherical-harmonic reduction → mixed-precision quantization) achieving a **74% mean size reduction** across three scenes at a 1.3 to 2.7 dB PSNR cost.

Diagnosed a per-column INT8 quantization collapse (~15 dB uniform) and recovered to within **0.1 to 0.7 dB of FP16** using Morton Z-order sub-group ranges, a positive and reproducible quantization finding.

Isolated a zero-update failure in post-pruning fine-tuning (stale optimizer state), reported as a negative result.

Ran the full multi-scene pipeline on a single NVIDIA T4 (Kaggle) using the Nerfstudio / gsplat ecosystem.

Rendering-aware SH vector quantization for compact 3DGS: a pre-registered replication and dissection · IN PREPARATION

Pre-registered study run from a gsplat fork, testing whether a rendering-derived codebook metric holds on held-out test views, with controlled fine-tuning ablations.

OPEN SOURCE CONTRIBUTIONS

gsplat PR #1063 (open): opacity × area weighted K-means for the `PngCompression` SH codebook, **+0.111 dB** mean PSNR on 9 MipNeRF360 scenes and +0.052 dB on Tanks & Temples, better on every scene at the same size and format.

gsplat PR #1061 (open): fixed `PngCompression` failing on `sh_degree=0` scenes and empty tensors.

PROJECTS

splat-slim — Python · Typer · NumPy · CI · **repo**

Training-free CLI / library that compresses trained 3DGS `.ply` files ~74% with <3 dB PSNR loss; ships self-decodable Morton-ordered sub-group INT8 with per-group ranges in a sidecar `.meta.npz`.

splat-lab — Vite · TypeScript · Three.js · **live demo**

Real-time in-browser demo of the pipeline with live sliders and an honest size readout; a TypeScript port of splat-slim.

nerf-prep — Python · Docker · GitHub Actions · **repo**

One-command Dockerized COLMAP → Nerfstudio prep: capture-aware matching, GPU/CPU auto-detect, registration report.

paper-chat — Python · Streamlit · sentence-transformers · **repo**

RAG over a PDF / arXiv library with per-(paper, page) citations; declines when its sources don't cover the question.

lumina-lms — Next.js · React · TypeScript · Prisma · **repo**

Full-stack AI learning management system: a retrieval-augmented tutor that answers from course materials and cites its sources, role-based access for students, instructors and admins, adaptive learning paths, gamification and analytics dashboards.

medical-expert-system — Python · Flask · scikit-learn · **repo**

Diagnosis system ensembling rule-based confidence factors, A* search, Bayesian inference and a Decision Tree + MLP.

TECHNICAL SKILLS

LANGUAGES	Python, C, C++, C#, TypeScript / JavaScript, Bash, LaTeX
ML & 3D VISION	3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, PyTorch, scikit-learn, COLMAP, model quantization, Bayesian inference, RAG & embeddings
TOOLS & INFRA	Next.js, React, Prisma, Flask, Streamlit, Three.js, Vite, Docker, Git, GitHub Actions (CI/CD), Linux

REFERENCES

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